

SMS Based HOME AUTOMATION Using Arduino

BISWAJIT DAS

Today, GSM networks are widely-used, whether for making calls or sending SMSes. In this project we will be controlling home appliances by sending SMSes.

There are many types of home automation systems available in the market. Most are simple home appliances-controlling systems like DTMF-controlled home appliances, IR remote controlled home appliances and radio frequency (RF) based home appliances. The main aim of this article is to develop an electronic application that uses GSM mobile technology to control different electronic/electrical equipment.

This system executes accurate equipment control according to the message sent by the mobile phone. Just by sending a message to the GSM modem interfaced to a microcontroller (MCU), you can perform on/off operations.

You can also verify the present status of the equipment through their display status on the LCD.

Circuit and working

Circuit of the SMS based home automation system using Arduino is shown in Fig. 1. It works off a 12V power supply (connected across CON1). An Arduino MCU requires only 5V but a GSM modem requires 12V. As Arduino Uno has an inbuilt 5V voltage regulator, a common 12V supply can be used for the whole system.

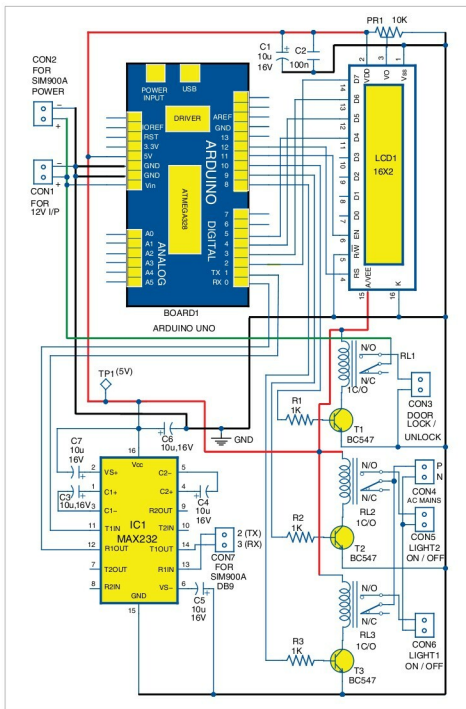


Fig. 1: Circuit diagram of the SMS based home automation using Arduino